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**Re: DCVRP in SWIFT**

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**From** Dingle, Clayton, M. (Clay) <clayton.dingle@spa.com>

**Date** Wed 4/15/2026 4:57 PM

**To** Waetjen, Tonia, N. (Tonia) <tonia.waetjen@spa.com>; Selke, Karl (Karl) <karl.selke@spa.com>; Bussard, Paul (Paul) <paul.bussard@spa.com>; Wise, Ben, P. (Ben) <ben.wise@spa.com>

Hi Ben,

After working through the plan with the team, I think it is in our best interest for you to use some J7 hours to complete the below:

To analyze the process of multi-hop transport, changing vehicles and/or domains, a very minimal, lightweight DESIM could be used. The basic DESIM functionality is written, and only a few events (pickup, dropoff, change location) will be needed: no modeling of combat, sensing, etc. is expected. A little DESIM instance is started. Basically, the process of building itineraries, then picking up, moving, and dropping off continues until either 5 days elapses or everything gets to its destination. This produces a record of where everything went, step by step, on which vehicles. Then the DESIM instance is deleted.

I will work on:

As Clay said, the first step is for a yet-to-be-written piece of Java code that produces a ScenarioRecord (as defined in the current DCVRP code). The key pieces of information of what resources are currently where, where they should go, what is their priority, what transport assets are where (changes with damage, redeployment, replacement, etc), what places they can access (changes with damage, repair, or whatever happened before this turn).

And:

Those records would go to a yet-to-be-written piece of Java code that would organize and display for the players information about what could be delivered. They could then see where they had logistical problems and where they did not. Because the DESIM instances are short-lived, it would be perfectly feasible to do this look ahead several times as the players juggle their priorities to try to do the best they can with limited resources.

I know you are still engaged with a separate effort, but if you think it is feasible to get something working by the first week of May, I think we will have enough time to prototype the remaining SWIFT integration for the May event. Does that work for you?

Let us know if you have any questions or concerns.

Thanks!

Clay

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**From:** Dingle, Clayton, M. (Clay) <clayton.dingle@spa.com>  
**Sent:** Monday, April 13, 2026 4:50 PM  
**To:** Waetjen, Tonia, N. (Tonia) <tonia.waetjen@spa.com>  
**Subject:** Fw: DCVRP in SWIFT

FYSA

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**From:** Wise, Ben, P. (Ben) <ben.wise@spa.com>  
**Sent:** Monday, April 13, 2026 1:48 PM  
**To:** Bussard, Paul (Paul) <paul.bussard@spa.com>  
**Cc:** Dingle, Clayton, M. (Clay) <clayton.dingle@spa.com>; Selke, Karl (Karl) <karl.selke@spa.com>; Wise, Ben, P. (Ben) <ben.wise@spa.com>  
**Subject:** Re: DCVRP in SWIFT

Paul,

Here's an outline of how it might go. The idea is to look at the current SWIFT state and say how much of what can be moved where over the next five days. The way to do that is a small embedded DESIM that does not interact with the rest of SWIFT in any way.

As Clay said, the first step is for a yet-to-be-written piece of Java code that produces a ScenarioRecord (as defined in the current DCVRP code). The key pieces of information of what resources are currently where, where they should go, what is their priority, what transport assets are where (changes with damage, redeployment, replacement, etc), what places they can access (changes with damage, repair, or whatever happened before this turn).

Then the DCVRP library will build a series of "itineraries" saying which items are carried by which transports where. They might get all the way to their final destination in one trip (which might last several days and stop at several locations, dropping off other stuff), or they might have to go to an intermediate node to change transports. A big plane might deliver stuff to a place where it gets put on a boat that can go to a beach where no airport is available. To analyze the process of multi-hop transport, changing vehicles and/or domains, a very minimal, lightweight DESIM could be used. The basic DESIM functionality is written, and only a few events (pickup, dropoff, change location) will be needed: no modeling of combat, sensing, etc. is expected. A little DESIM instance is started. Basically, the process of building itineraries, then picking up, moving, and dropping off continues until either 5 days elapses or everything gets to its destination. This produces a record of where everything went, step by step, on which vehicles. Then the DESIM instance is deleted.

Those records would go to a yet-to-be-written piece of Java code that would organize and display for the players information about what could be delivered. They could then see where they had logistical problems and where they did not. Because the DESIM instances are short-lived, it would be perfectly feasible to do this look ahead several times as the players juggle their priorities to try to do the best they can with limited resources.

Those two new pieces of software are the "adapter" between DCVRP and SWIFT.

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**From:** Bussard, Paul (Paul) <paul.bussard@spa.com>

**Sent:** Monday, April 13, 2026 1:36 PM

**To:** Wise, Ben, P. (Ben) <ben.wise@spa.com>

**Cc:** Dingle, Clayton, M. (Clay) <clayton.dingle@spa.com>; Selke, Karl (Karl) <karl.selke@spa.com>

**Subject:** DCVRP in SWIFT

The adapter for DCVRP into SWIFT for JSWAM-O isn't fully clear to me; there seems to be no way to execute the DCVRP math and functions from within SWIFT even if the objects are translatable. Can you help me understand what's necessary?

Thanks,  
Paul B.



Paul Bussard | Wargame Designer and Analyst

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